

HireKarma Pvt. Ltd.

Full Stack Engineering Assessment – SDE-1

Thank you for applying for the Software Development Engineer (SDE-1) role at HireKarma Pvt. Ltd.

Due to the large number of applications, we have designed this Full Stack Development Assessment to evaluate your practical engineering skills. This assignment will help us understand your ability to design, build, and deploy a real-world application.

Skills Evaluated

- Authentication & Security
- REST API Development
- Database Design
- Web Scraping
- Frontend Development
- Deployment
- Problem Solving
- Code Quality

Assignment: Job Scraper Platform

You are required to build a Job Scraper Web Application where users can log in and scrape job listings from multiple platforms.

1. Authentication System

- User Signup
- User Login
- JWT or token-based authentication
- Protected routes (only logged-in users can access the platform)

2. Job Scraper Module

- User enters job keyword
- User enters location
- Click 'Scrape Jobs' button
- Scrape jobs from platforms such as LinkedIn, Naukri, Internshala, and Unstop

3. Job Results Table

- Job Title
- Company Name
- Location
- Job Platform

- Job Link
- Apply Button

4. Apply to Job

- Click Apply button to open the job link
- Store the application record in the database
- Record Job Title, Company, Platform, Date Applied, and Status

5. Application History

- Display list of jobs applied by the user
- Show platform, applied date, and job link
- Maintain application history in database

6. Profile Page

- View user profile
- Update profile details
- Delete Image
- Demonstrate CRUD operations

Technical Requirements

Frontend (Preferred):

- Next.js
- TypeScript
- Tailwind CSS

Backend (Preferred):

- Python (FastAPI)
- REST APIs

Database:

- PostgreSQL or any relational DB

Version Control:

- Git + GitHub

Deployment (Mandatory)

- Frontend deployed on Vercel, Netlify, or similar
- Backend deployed on AWS, Render, Railway, or similar
- Application must be publicly accessible

AI Integration (Bonus Points)

Candidates will receive additional preference if they integrate AI-related features.

- AI job recommendation
- Resume matching with job description
- AI summary of job descriptions
- Smart job filtering

Submission Requirements

- Public GitHub Repository Link
- Live Frontend URL
- Backend API URL
- README file explaining the project

Evaluation Criteria

- Working application
- Code quality and architecture
- Git usage and commit history
- Successful deployment
- UI/UX quality
- Security practices

Optional Bonus Features

- Pagination
- Filters
- Bookmark jobs
- Resume upload
- Email notifications